1 The estimation problem

Classical control theory generally assumes that the system model is known exactly and that measurements are error-free. Under these assumptions, let $x(t)$ denote the system state and $u(t)$ the applied control input. The dynamics are described by

$$\begin{aligned} \dot{x}(t) &= f(x(t), u(t)) \\ y(t) &= h(x(t), u(t)) \end{aligned}$$

In practice, these assumptions rarely hold. The mathematical model only approximates the actual system and may be affected by uncertain parameters, unmodelled dynamics, neglected nonlinearities, and external disturbances. Sensor measurements are also generally contaminated by noise.

Let $\xi(t)$ denote the process noise, which accounts for uncertainties and dynamics that the model does not fully capture, and $\eta(t)$ the measurement noise. A more realistic representation of the system is

$$\begin{aligned} \dot{x}(t) &= f(x(t), u(t)) + \xi(t) \\ y(t) &= h(x(t), u(t)) + \eta(t) \\ x(0) &= x_0 + \xi_0 \end{aligned}$$

The system state generally cannot be measured directly, so estimation algorithms reconstruct it from the available measurements and the system model.

The main estimation techniques fall into three categories.

- **Smoothing**. The state at time t is estimated using measurements obtained both before and after time t. Since smoothing requires future information, it can only reconstruct past states and is unsuitable for real-time estimation of the current state.

- **Filtering**. The state at time t is estimated using only the measurements available up to that time. This is the most common technique in real-time control applications.

- **Prediction**. The state at a future time is estimated using only information available up to an earlier time t_1, with $t_1 < t$.

2 The Kalman filter

The **Kalman filter** is a widely used tool for state estimation when the system is LTI and has the form

$$\begin{aligned} \dot{x}(t) &= A\,x(t) + B\,u(t) + \xi(t) \\ y(t) &= C\,x(t) + \eta(t) \\ x(0) &= x_0 + \eta_0 \end{aligned}$$

The algorithm makes the following assumptions.

- The process noise $\xi(t)$ and measurement noise $\eta(t)$ both have zero mean.

$$E[\xi(t)] = 0 \qquad E[\eta(t)] = 0$$

- The noise at each instant is independent of its past values.
- The process noise, measurement noise, and initial error are mutually uncorrelated.

$$E[\xi(t)\,\eta^T(t)] = 0 \qquad E[\xi(t)\,x^T(0)] = 0 \qquad E[\xi(t)\,x^T(0)] = 0$$

- The initial covariance is given by

$$E[\xi_0 \xi_0^T] = P_0$$

- The noise processes are Gaussian.

To minimise the mean-square error, minimise the cost functional

$$J = \frac{1}{2}(x(0) - x_0)^T P_0^{-1}(x(0) - x_0) + \frac{1}{2}\int_0^{t_f} (\dot{x} - A\,x)^T R(t)\,(\dot{x} - A\,x) + (y - C\,x)^T Q(t)(y - C\,x)\mathrm{dt}$$

where

- the term $\frac{1}{2}(x(0) - x_0)^T P_0^{-1}(x(0) - x_0)$ penalises deviations from the initial estimate x_0
- the term $\frac{1}{2}\int_0^{t_f} (\dot{x} - A\,x)^T\,R(t)\,(\dot{x} - A\,x)\,\mathrm{dt}$ penalises deviations from the nominal model $\dot{x} = A\,x$
- the term $\frac{1}{2}\int_0^{t_f} (y - C\,x)^T Q(t)(y - C\,x)\mathrm{dt}$ penalises discrepancies between the observed and predicted measurements.

Minimising J can be formulated as an optimal control problem. Defining

$$u(t) = \dot{x} - A\,x$$

gives the following cost functional.

$$J = \frac{1}{2}(x(0) - x_0)^T P_0^{-1}(x(0) - x_0) + \frac{1}{2}\int_0^{t_f} u(t)^T R(t)\,u(t) + (y - C\,x)^T Q(t)(y - C\,x)\mathrm{dt}$$

The cost is minimised when

$$\dot{P} = A\,P + P A + R^{-1} - P C^T Q\,C\,P$$

The Kalman filter equations then become

$$\begin{aligned} \dot{\hat{x}} &= A\,\hat{x} + K(t)(y - C\,\hat{x}) \\ \dot{P} &= A\,P + P A + R^{-1} - K(t) C\,P \\ K(t) &= P(t)\,C^T Q \end{aligned}$$

3 Assignment 8

Consider the model of a continuous stirred-tank reactor

$$\begin{aligned} \dot{x}_1 &= -(1+D\,a_1)x_1 + u_1 + \xi_1 \\ \dot{x}_2 &= D\,a_1\,x_1 - (1+D\,a_2)\,x_2 + u_2 + \xi_2 \\ y &= x_2 + \eta \\ x_1(0) &= x_{10} + \xi_{10} \\ x_2(0) &= x_{20} + \xi_{20} \end{aligned}$$

where

- x_1 is the concentration of species A
- x_2 is the concentration of species B
- the reactions are $A \to A$, $A \to B$, and $B \to B$
- $u_i, i = 1, 2$ denotes the feeds of species A and B
- D is the dilution coefficient
- a_1 and a_2 are the rate constants for the first-order reactions.

The variables

$$\xi = \begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix}$$

and η represent process noise and measurement noise, respectively. Both are assumed to be white, Gaussian, and zero-mean at all times.

Also assume that

- the process noise covariance is

$$R^{-1} = 0$$

- the measurement noise covariance is

$$Q^{-1} = 0.0025$$

- the initial error covariance is

$$P_0 = \begin{pmatrix} 0.04 & 0 \\ 0 & 0.01 \end{pmatrix}$$

Finally, assume that

- $D\,a_1 = 3$
- $D\,a_2 = 1$
- $t_f = 2$
- $x_{10} = 1$

- $x_{20} = 0$
- $\hat{x}_1(0) = 0.85$
- $\hat{x}_2(0) = 0.15$
- $u_1 = u_2 = 0$

Generate the noise signals using random number generators and complete the following tasks.

1. Show that the system is observable.
2. Write out the filtering Riccati differential equation (FDRE) for this problem and determine its initial condition.
3. Integrate the FDRE and plot the evolution of its entries over time.
4. Simulate the system and the filter, then plot the state variables and their estimates over time.
5. Repeat Step 4 with measurements sampled at period

   $$t_s = 0.05,$$

   and compare the results with those of Step 4, explaining the observed differences.

4 Solution

4.1 Observability check

First, write the system in standard form. We obtain

$$\begin{aligned} \dot{x}_1 &= -(1+D\,a_1)x_1 + u_1 + \xi_1 \\ \dot{x}_2 &= D\,a_1\,x_1 - (1+D\,a_2)\,x_2 + u_2 + \xi_2 \\ y &= x_2 + \eta \end{aligned}$$

$$\begin{aligned} \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} &= \begin{pmatrix} -(1+D\,a_1) & 0 \\ D\,a_1 & -(1+D\,a_2) \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + I \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} + I \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} \\ y &= \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \eta \end{aligned}$$

The system is therefore described by

$$\begin{aligned} \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} &= \begin{pmatrix} -4 & 0 \\ 3 & -2 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + I \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} + I \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} \\ y &= \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \eta \end{aligned}$$

Next, compute the observability matrix $\mathcal{O}$. We obtain

$$\begin{aligned} \mathcal{O} &= \begin{pmatrix} C \\ C\,A \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ 3 & -2 \end{pmatrix} \end{aligned}$$

Since $\mathrm{rk}(\mathcal{O}) = 2$, the system is observable, so all states can be estimated from the available measurements.

4.2 Filtering Riccati differential equation

The filtering Riccati differential equation is

$$\dot{P} = A\,P + P\,A + R^{-1} - P(t)\,C^T\,\mathrm{QC}\,P$$

where $P = P^T \geq 0$. Defining

$$P = \begin{pmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{pmatrix}$$

gives

$$\begin{aligned} A\,P + P\,A^T + R^{-1} - P\,C^T\,\mathrm{QC}\,P &= \begin{pmatrix} -4 & 0 \\ 3 & -2 \end{pmatrix} \cdot \begin{pmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{pmatrix} + \begin{pmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{pmatrix} \cdot \begin{pmatrix} -4 & 3 \\ 0 & -2 \end{pmatrix} + 0 - \\ & \quad \begin{pmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} \cdot \frac{1}{0.0025} \cdot \begin{pmatrix} 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{pmatrix} \\ &= \begin{pmatrix} -4p_{11} & -4p_{12} \\ 3p_{11} - 2p_{12} & 3p_{12} - 2p_{22} \end{pmatrix} + \begin{pmatrix} -4p_{11} & 3p_{11} - 2p_{12} \\ -4p_{12} & 3p_{12} - 2p_{22} \end{pmatrix} - \\ & \quad 400 \cdot \begin{pmatrix} p_{12} \\ p_{22} \end{pmatrix} \cdot \begin{pmatrix} p_{12} & p_{22} \end{pmatrix} \\ &= \begin{pmatrix} -8p_{11} & 3p_{11} - 6p_{12} \\ 3p_{11} - 6p_{12} & 6p_{12} - 4p_{22} \end{pmatrix} - 400 \cdot \begin{pmatrix} p_{12} \\ p_{22} \end{pmatrix} \cdot \begin{pmatrix} p_{12} & p_{22} \end{pmatrix} \end{aligned}$$

$$= \begin{pmatrix} -8p_{11} & 3p_{11} - 6p_{12} \\ 3p_{11} - 6p_{12} & 6p_{12} - 4p_{22} \end{pmatrix} - 400 \cdot \begin{pmatrix} p_{11}^2 & p_{12}p_{22} \\ p_{12}p_{22} & p_{22}^2 \end{pmatrix}$$
$$= \begin{pmatrix} -8p_{11} - 400p_{12}^2 & 3p_{11} - 6p_{12} - 400p_{12}p_{22} \\ 3p_{11} - 6p_{12} - 400p_{12}p_{22} & 6p_{12} - 4p_{22} - 400p_{22}^2 \end{pmatrix}$$

It follows that

$$\begin{aligned} \dot{p}_{11} &= -8p_{11} - 400p_{12}^2 \\ \dot{p}_{12} &= 3p_{11} - 6p_{12} - 400p_{12}p_{22} \\ \dot{p}_{22} &= 6p_{12} - 4p_{22} - 400p_{22}^2 \end{aligned}$$

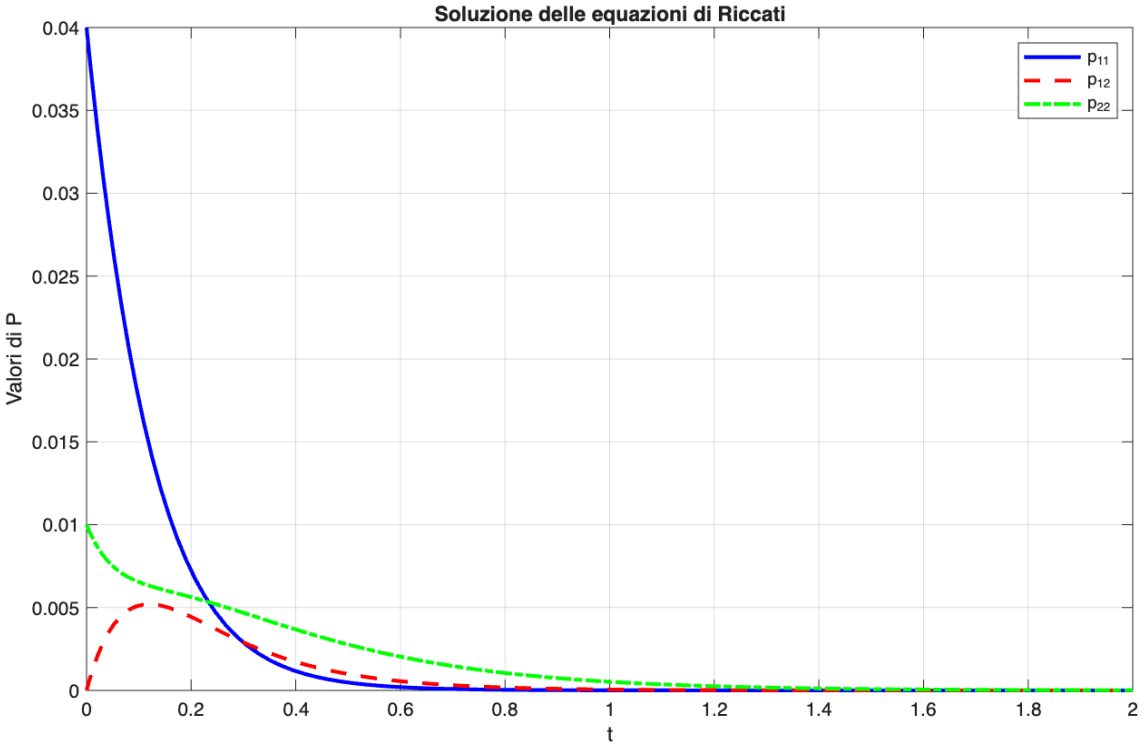
Since

$$P_0 = \begin{pmatrix} 0.04 & 0 \\ 0 & 0.01 \end{pmatrix}$$

by assumption,

$$\begin{aligned} p_{11}(0) &= 0.04 \\ p_{12}(0) &= 0 \\ p_{22}(0) &= 0.01 \end{aligned}$$

The evolution of the matrix P over the interval $[0, 2]$ is shown below.

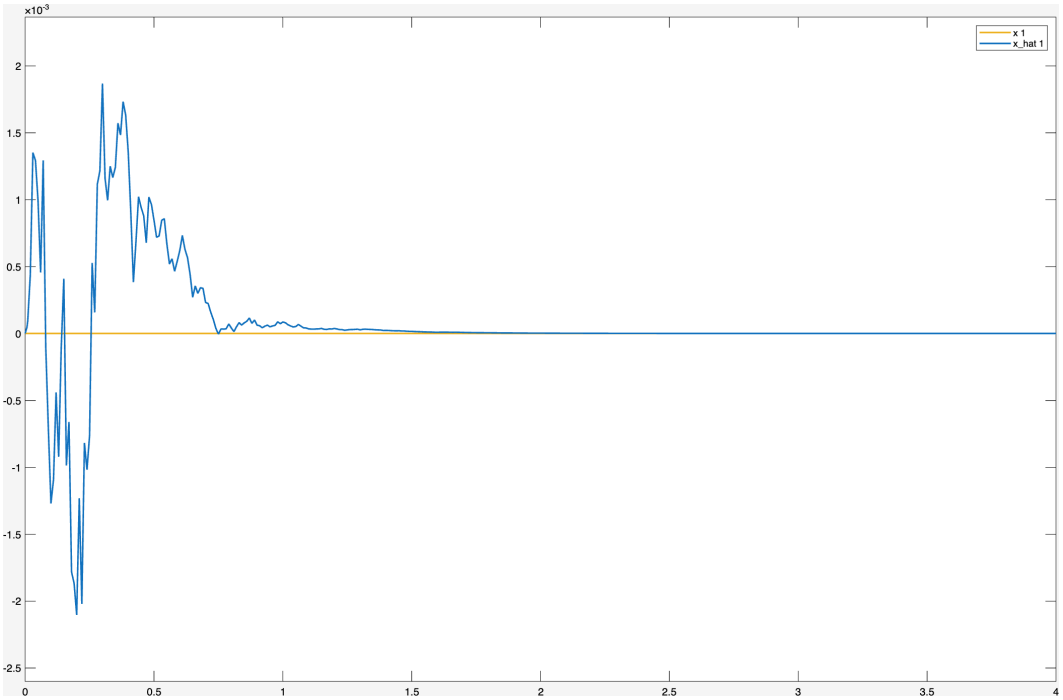


5 State estimation

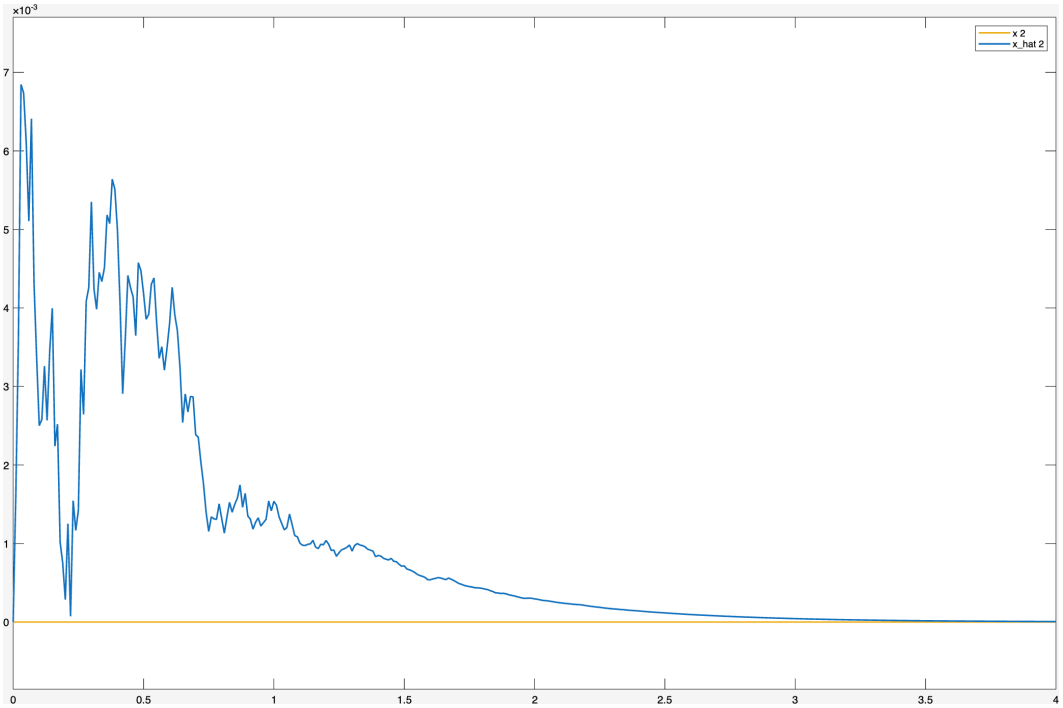
5.1 Continuous measurements

With continuous output measurements, the results are as follows.

- The estimate $\hat{x}_1$ of state x_1 converges in about $2s$ and reaches a maximum amplitude of approximately 2.



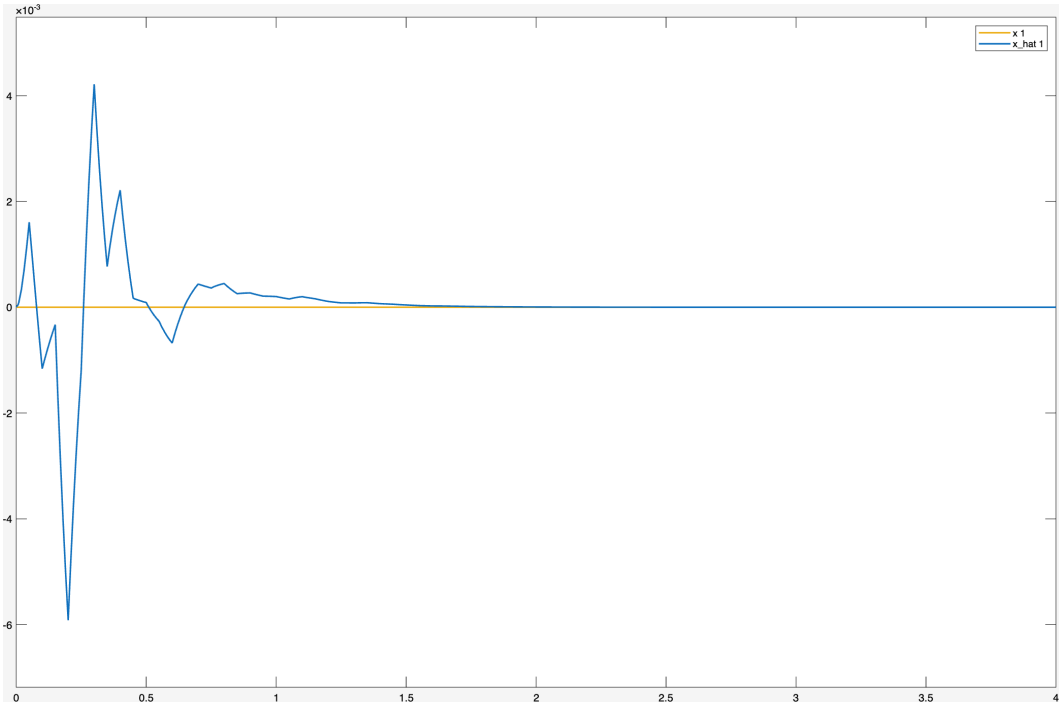
- The estimate $\hat{x}_2$ of state x_2 converges in about $3.6s$ and reaches a maximum amplitude of approximately 7.



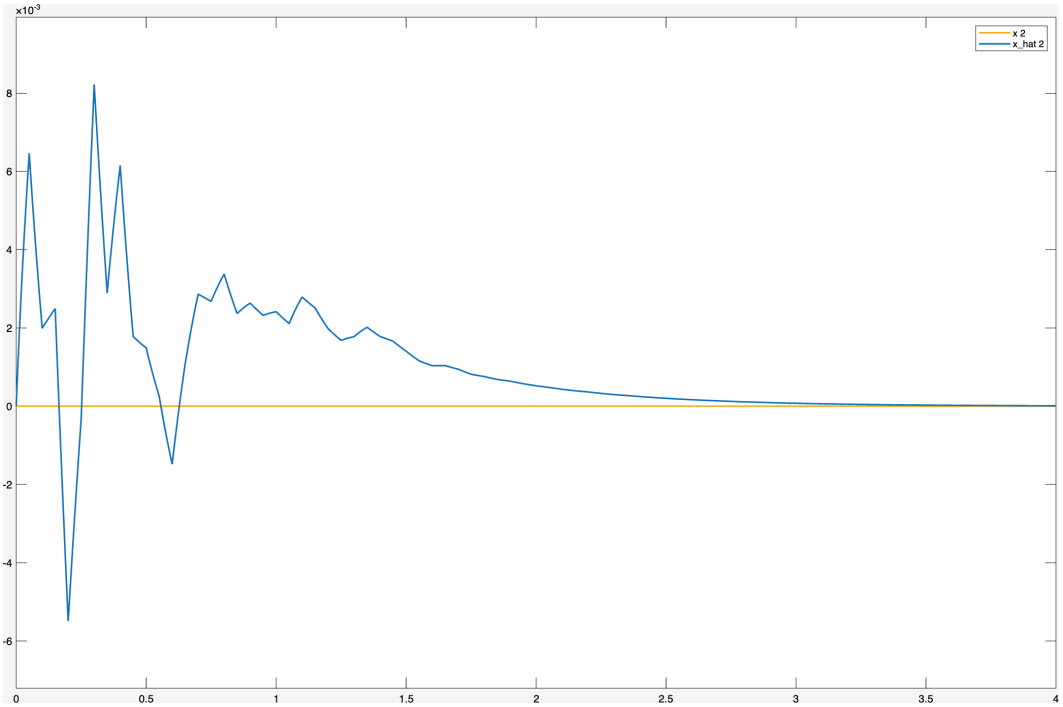
5.2 Sampled measurements

With sampled output measurements, the results are as follows.

- The estimate $\hat{x}_1$ of state x_1 converges in about $2s$ and reaches a maximum amplitude of approximately $4 \cdot 10^{-3}$.



- The estimate $\hat{x}_2$ of state x_2 converges in about $3.6s$ and reaches a maximum amplitude of approximately $8 \cdot 10^{-3}$.



Sampling the measurement introduces a zero-order hold, so the estimator receives a new measurement every $T_s = 0.05\ s$. The observer's correction term is consequently updated less frequently. In this simulation, the estimates exhibit a more damped response and smaller peaks.